

EVAPORATION

Group1

Mahamat Abdraman

Yohannes Teweldeberhan

Gewargis Garmooos

Sinan Masoud

Marvl Odeesh

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Introduction

- ❑ The purpose of this lab is to observe the effect of vacuum pressure on economy and capacity over a range of pressures for a single effect evaporator and analyze the underlying parameters of the system.
- ❑ Vacuum pressures set at 10, 15 and 20 in inHg will be used in each trial
- ❑ Having the evaporator to reach steady state with stable temperature and volume is integral part of the experiment

Theory

- ❑ Evaporation is a process that acts by vaporizing a part of the volatile solvent concentrating the solution. Thus, evaporation is basically governed by the principles of heat transfer
- ❑ The overall assumption is that the system is in steady state
- ❑ Governing Equations:
 - ❑ Heat Transfer Rate: $q = UA\Delta T$
 - ❑ Economy = amount of product(kg)/time(hr)
 - ❑ Capacity = kg of product/kg of stream used
 - ❑ Steam Consumption = Economy/Capacity
- ❑ Connection to the Experiment
 - ❑ Energy Balances with two control volumes (i.e.: Steam trap)

Apparatus

- ☐ Evaporator Vessel
- ☐ Steam Trap
- ☐ Make-up Water Tank
- ☐ Thermocouple Digital Monitor
- ☐ Steam Trap Erlenmeyer Flask
- ☐ Condensate Erlenmeyer Flasks

Material & Supplies

- ☐ Graduated cylinder(1000 mL)
- ☐ Steam
- ☐ Water
- ☐ Sugar
- ☐ Electricity
- ☐ Time Measuring Device

Procedure

Prelab Preparations

- ☐ Pick a reasonable marking on the evaporator vessel and measure the amount of water it takes to reach that mark.
- ☐ From the amount of water measured, calculate how many grams of sugar would be needed to make 10% weight solution
- ☐ Perform a quick run with water to see if there is still any residual liquid in the container or whether any expansion in fluid level may arise

Procedure

- ☐ Fill the evaporator vessel with 10% weight solution
- ☐ Turn on the steam, the cooling water, and vacuum pump
- ☐ Once the fluid level in the container change, adjust the make-up water valve to bring fluid level and concentration to steady state
- ☐ Collect steam and condensate before reaching steady state
- ☐ Once steady state is reached, switch the valves to collect condensate in the other Erlenmeyer flask
- ☐ Observe the amount of condensate and steam collected during steady state for 15 minutes
- ☐ Change to another vacuum pressure and repeat

Experimental Challenges

- ❑ The fluid level can be hard to stabilize at times with the valve and requires some fine control
- ❑ Low vacuum pressures create a longer time to reach steady state
- ❑ At times, the temperature readings of the thermocouple may fluctuate, wait about 10 minutes and use judgement to tell if it is due to not being at steady state or being an error on the instrument

Safety

- ❑ Collected steam and condensate can be very hot. Thus, wear appropriate gloves
- ❑ Steam lines are very hot as well. So, avoid touching anywhere steam passes through
- ❑ Parts use electricity and this is a wet lab; there can be a possibility for electrical hazards. Be aware of water spill